

专题 13. 定积分几何应用的解题方法.

$$1 + \cos \theta = 3 \cos \theta$$

$$\cos \theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}$$

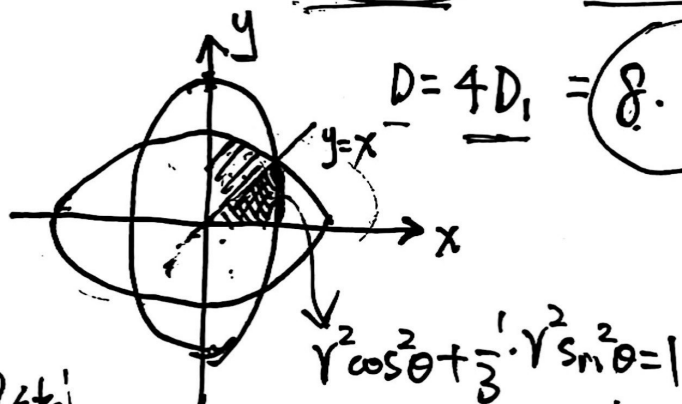
考法一. 平面图形的面积.

$$S = \frac{1}{2} \int_a^b r^2 \omega d\theta$$

(一). 有限区域 (基础题讲得不少, 这里只讲2个题). $\tan \theta$, $\arctan \frac{\sqrt{3}}{3} = ?$

例1. 求两椭圆 $x^2 + \frac{1}{3}y^2 = 1$ 和 $\frac{1}{3}x^2 + y^2 = 1$ 所围成的公共部分 D 的面积. (Ans: $\frac{2\sqrt{3}}{3}\pi$).

[解]:



$$D = 4D_1 = 8 \cdot \frac{1}{2} \int_0^{\frac{\pi}{4}} \frac{1}{\cos^2 \theta + \frac{1}{3} \sin^2 \theta} d\theta = 4 \cdot 3 \cdot \int_0^{\frac{\pi}{4}} \frac{1}{3 \cos^2 \theta + \sin^2 \theta} d\theta \cdot \frac{\sqrt{3}}{3}$$

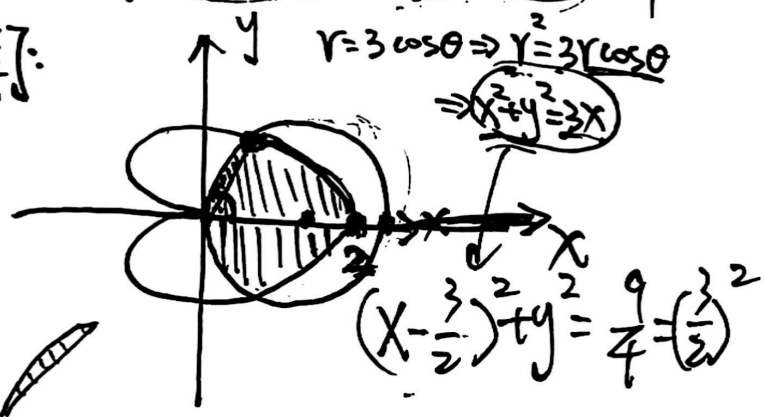
$$= 12 \cdot \int_0^{\frac{\pi}{4}} \frac{1}{3 + \tan^2 \theta} d \tan \theta = 12 \cdot \frac{1}{\sqrt{3}} \cdot \arctan \left(\frac{\tan \theta}{\sqrt{3}} \right) \Big|_0^{\frac{\pi}{4}}$$

$$= 4\sqrt{3} \cdot \left[\frac{\pi}{6} - 0 \right] = \frac{2\sqrt{3}}{3} \pi \checkmark$$

心形线.

例2. 求 $r = 1 + \cos \theta$ 和 $r = 3 \cos \theta$ 所围成的公共部分 D 的面积. (Ans: $\frac{5}{4}\pi$).

[解]:

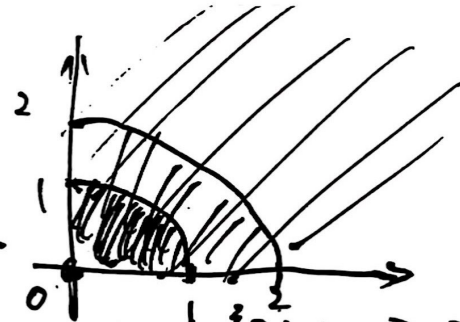


$$S = 2[S_H] = 2 \cdot \left[\frac{1}{2} \int_0^{\frac{\pi}{3}} (1 + \cos \theta)^2 d\theta + \frac{1}{2} \int_{\frac{\pi}{3}}^{\pi} 9 \cos^2 \theta d\theta \right]$$

$$= \frac{5}{4} \pi$$

(=). 无限正域.

例3. 设 $x \geq 0$. $f(x) = \lim_{t \rightarrow +\infty} (1 - \frac{x}{t})^{xt}$, $g(x) = \int_0^x f(u) du$. ★★



(1) 求 $y = g(x)$ 在 $x \geq 0$ 部分的水平渐近线; (2) 求 $y = g(x)$ 与其水平渐近线及 y 轴在 $x \geq 0$ 部分所围图形的面积 A .

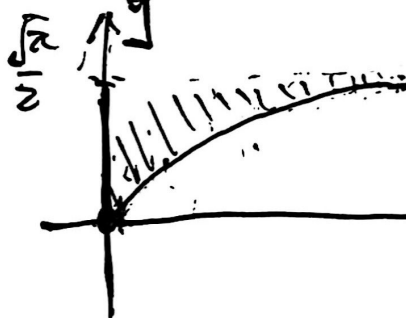
解: (1). $f(x) = e^{\lim_{t \rightarrow +\infty} \frac{1}{t} \cdot xt \cdot \ln(1 - \frac{x}{t})} = e^{-x^2} \Rightarrow g(x) = \int_0^x e^{-u^2} du$.

$x \geq 0$ 时. 水平渐近线. $\lim_{x \rightarrow +\infty} I = \int_0^{+\infty} e^{-x^2} dx = ???$ $I^2 = (\int_0^{+\infty} e^{-x^2} dx)^2 = \int_0^{+\infty} e^{-x^2} dx \cdot \int_0^{+\infty} e^{-y^2} dy$
 $= \int_0^{+\infty} \int_0^{+\infty} e^{-(x^2+y^2)} dx dy = \int_0^{\frac{\pi}{2}} d\theta \int_0^{+\infty} e^{-r^2} \cdot r dr = \frac{\pi}{2} \cdot (-\frac{1}{2}) \cdot e^{-r^2} \Big|_0^{+\infty} = (-\frac{\pi}{4}) \cdot (0 - 1) = \frac{\pi}{4}$.

$\Rightarrow I = \sqrt{\frac{\pi}{4}} = \frac{\sqrt{\pi}}{2}$ ★★ $\Rightarrow x \geq 0$ 时. $g(x)$ 的水平渐近线为 $y = \frac{\sqrt{\pi}}{2}$ ✓

(2) $g'(x) = e^{-x^2} > 0 \Rightarrow g(x) \uparrow \Rightarrow A = \int_0^{+\infty} [\frac{\sqrt{\pi}}{2} - \int_0^x e^{-u^2} du] dx = \frac{\sqrt{\pi}}{2} \int_0^{+\infty} 1 dx - \int_0^{+\infty} \int_0^x e^{-u^2} du dx$

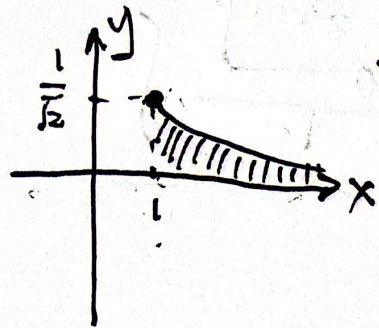
$= \lim_{x \rightarrow +\infty} \frac{\sqrt{\pi}}{2} x - \int_0^{+\infty} \int_0^x e^{-u^2} du dx = \lim_{x \rightarrow +\infty} \frac{\sqrt{\pi}}{2} x - \int_0^{+\infty} \frac{x^2}{2} e^{-x^2} dx + (-\frac{1}{2}) \cdot (0 - 1) = \lim_{x \rightarrow +\infty} \frac{x^2}{2} e^{-x^2} + \frac{1}{2} = \frac{1}{2}$ ✓



例4. (2023年). 已知平面区域 $D = \{(x, y) \mid 0 \leq y \leq \frac{1}{x \cdot \sqrt{1+x^2}}, x \geq 1\}$

(1) 求 D 的面积; (2). 求 D 绕 x 轴旋转所成的旋转体体积.

[解]: (1)



$$\Rightarrow S_D = \int_1^{+\infty} \frac{1}{x \cdot \sqrt{1+x^2}} dx = \int_1^{+\infty} \frac{1}{x^2 \sqrt{1+(\frac{1}{x})^2}} dx = - \int_1^{+\infty} \frac{1}{\sqrt{1+(\frac{1}{x})^2}} d\frac{1}{x}$$

$$= \ln \left[\frac{1}{x} + \sqrt{1+(\frac{1}{x})^2} \right] \Big|_1^{+\infty} = - \ln \frac{1}{1+\sqrt{2}} = \ln(1+\sqrt{2}) \quad \checkmark$$

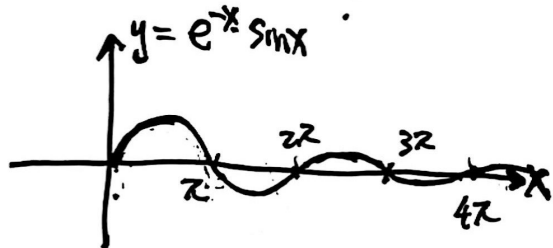
$$(2). V_x = \pi \cdot \int_1^{+\infty} y^2 dx = \pi \cdot \int_1^{+\infty} \frac{(1+x^2) - x^2}{x^2(1+x^2)} dx = \pi \cdot \int_1^{+\infty} \left(\frac{1}{x^2} - \frac{1}{1+x^2} \right) dx$$

$$= \pi \cdot \left[-\frac{1}{x} + \arctan x \right] \Big|_1^{+\infty} = (-\pi) \cdot \left[(0 + \frac{\pi}{2}) - (1 + \frac{\pi}{4}) \right]$$

$$= (-\pi) \cdot (\frac{\pi}{4} - 1) = \pi \cdot (1 - \frac{\pi}{4}) \quad \checkmark$$

例5. 求曲线 $y = e^{-x} \sin x$ ($x \geq 0$) 与 x 轴之间图形的面积. (Ans: $\frac{1}{2} \cdot \frac{e^{\pi} + 1}{e^{\pi} - 1}$).

[解]: ($x > 0$ 时, $0 < e^{-x} < 1$. 所以可以将 e^{-x} 看成函数 $y = \sin x$ 的“压缩系数”. 且 x 越大, 压缩越扁)



(草图)

$$\Rightarrow S = \int_0^{+\infty} |y| dx = \int_0^{+\infty} e^{-x} |\sin x| dx. \text{ 由于 } \int_0^{+\infty} e^{-x} |\sin x| dx \leq \int_0^{+\infty} e^{-x} dx = 1.$$

由比较判别法, $\int_0^{+\infty} e^{-x} |\sin x| dx$ 收敛! 此时才有 $S = \lim_{n \rightarrow \infty} \int_0^{n\pi} e^{-x} |\sin x| dx$

$$= \lim_{n \rightarrow \infty} \sum_{k=1}^n \int_{(k-1)\pi}^{k\pi} e^{-x} (-1)^{k-1} \sin x dx = \lim_{n \rightarrow \infty} \sum_{k=1}^n (-1)^{k-1} \int_{(k-1)\pi}^{k\pi} e^{-x} \sin x dx$$

$$\text{又由 } \int_{(k-1)\pi}^{k\pi} e^{-x} \sin x dx = -\frac{e^{-x}}{2} (\sin x + \cos x) \Big|_{(k-1)\pi}^{k\pi} = -\frac{e^{-k\pi}}{2} (0 + (-1)^k) + \frac{e^{-(k-1)\pi}}{2} (0 + (-1)^{k-1})$$

$$= \frac{1}{2} e^{-(k-1)\pi} (-1)^{k-1} [e^{-\pi} + 1] \Rightarrow \int_0^{+\infty} e^{-x} |\sin x| dx = \frac{e^{-\pi} + 1}{2} \lim_{n \rightarrow \infty} \sum_{k=1}^n e^{-(k-1)\pi}$$

$$= \frac{e^{-\pi} + 1}{2} \lim_{n \rightarrow \infty} \sum_{k=1}^n (e^{-\pi})^{k-1} = \frac{e^{-\pi} + 1}{2} \lim_{n \rightarrow \infty} \frac{1 - (e^{-\pi})^n}{1 - e^{-\pi}} = \frac{e^{-\pi} + 1}{2} \cdot \frac{1}{1 - e^{-\pi}}$$

$$= \frac{1}{2} \cdot \frac{e^{\pi} + 1}{e^{\pi} - 1} \checkmark$$

[易解]: $I = \int_0^{+\infty} e^{-x} \cdot (\sin x) dx \xrightarrow{x+\tau=t} \int_\tau^{+\infty} e^{-(t-\tau)} \cdot |\sin t| dt \cdot e^{a+b} = e^a \cdot e^b$

$\tau = \tau.$

$= e^\tau \cdot \int_\tau^{+\infty} e^{-t} |\sin t| dt = \underline{(e^\tau)} \left[I - \int_0^\tau e^{-x} \sin x dx \right].$

$\Rightarrow (e^\tau - 1) \cdot I = e^\tau \cdot \int_0^\tau e^{-x} \sin x dx \Rightarrow I = \frac{1}{2} \cdot \frac{e^\tau + 1}{e^\tau - 1} \checkmark.$



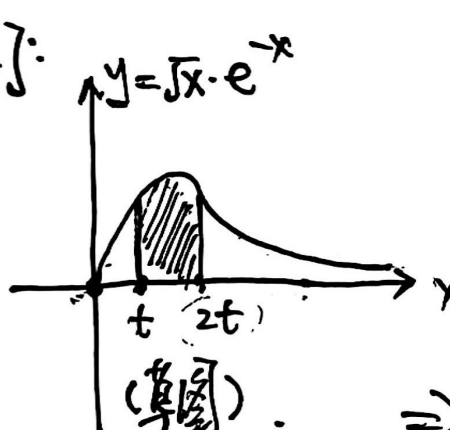
线上**5分/页**打印，**包邮**到家
操作简单，一看就会

考法二：旋转体的体积

(一) 绕水平轴或竖直轴旋转

$$y(0)=0, \quad y(+\infty)=0, \quad y(x)>0.$$

例 6. (2024 年). 设 $t>0$. 平面区域 D 由曲线 $y=\sqrt{x}e^{-x}$ 与直线 $x=t$, $x=2t$ 及 x 轴围成.
 D 绕 x 轴旋转一周所成旋转体的体积为 $V(t)$. 求 $V(t)$ 的最大值. (Ans: $\frac{\ln^2}{16}\pi + \frac{3}{64}\pi$).

解:  $y=\sqrt{x} \cdot e^{-x}$

$$V(t) = \pi \int_t^{2t} x \cdot e^{-2x} dx \quad (t>0)$$
$$V'(t) = \pi \cdot \left[2 \cdot 2t e^{-4t} - t e^{-2t} \right] = \pi t \cdot e^{-2t} \cdot [4e^{-2t} - 1]$$

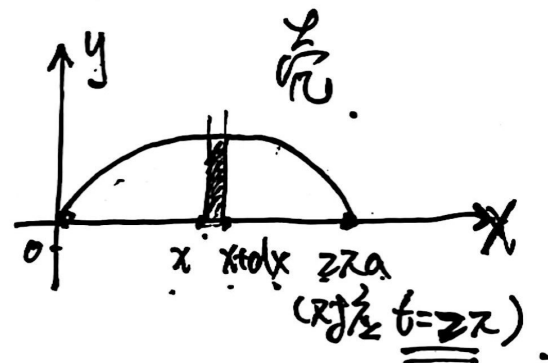
(草图). $\Rightarrow V(t)$ 在 $(0, \ln^2)$ $\uparrow$, 在 $(\ln^2, +\infty)$ $\downarrow$

$$\Rightarrow [V(t)]_{\max} = V(\ln^2) = \frac{\ln^2}{16} \cdot \pi + \frac{3}{64} \cdot \pi \quad \checkmark$$

例7. 摆线 $L: \begin{cases} x = a(t - \sin t) \\ y = a(1 - \cos t) \end{cases} \quad (0 \leq t \leq 2\pi, a > 0)$. 设 L 与 x 轴围成的图形为 D .

求 D 绕着 x 轴旋转一周而成的旋转体体积 V_x

y 轴. $\dots\dots\dots V_y$



[解]: (1) $V_x = \pi \int_0^{2\pi a} y^2 dx = \pi \int_0^{2\pi} a^2(1 - \cos t)^2 \cdot a(1 - \cos t)' dt = \pi a^3 \int_0^{2\pi} (1 - \cos t)^3 dt$.

$= 2\pi a^3 \int_0^{2\pi} \left(2 \sin^2 \frac{t}{2}\right)^3 dt = 2\pi a^3 \cdot 8 \cdot \int_0^{\pi} \sin^6 u du = 2\pi a^3 \cdot 8 \cdot 2 \cdot \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} = \dots$

(2) $V_y = \int_0^{2\pi a} (2\pi \cdot x) y dx = 2\pi \int_0^{2\pi} a(t - \sin t) \cdot a(1 - \cos t) \cdot a(1 - \cos t) dt$. ↑ 偶

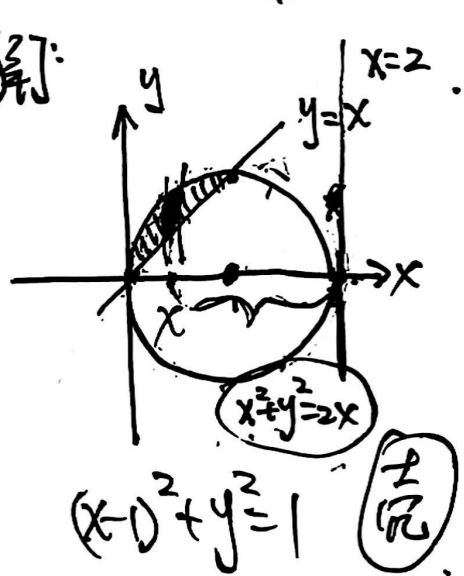
$= 2\pi a^3 \int_0^{2\pi} (t - \sin t) \cdot (1 - \cos t)^2 dt \xrightarrow{t - \pi = u} 2\pi a^3 \int_{-\pi}^{\pi} [\pi + u + \sin u] (1 + \cos u)^2 du$

$= 2\pi^2 a^3 \cdot 2 \cdot \int_0^{\pi} (1 + \cos u)^2 du = 4\pi^2 a^3 \cdot 2 \int_0^{\pi} \left(2 \cos^2 \frac{u}{2}\right) d\left(\frac{u}{2}\right) = 8\pi^2 a^3 \cdot 4 \int_0^{\frac{\pi}{2}} \cos^4 p dp$

$= 32\pi^2 a^3 \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} = \dots$

例8 求由 ~~曲线~~ $x^2+y^2 \leq 2x$ 与 $y \geq x$ 确定的平面图形绕直线 $x=2$ 旋转而成的体积. (Ans: $\frac{\pi^2}{2} - \frac{2}{3}\pi$)

[解]:



$$V = \int_0^1 2\pi(2-x) \left(\sqrt{2x-x^2} - x \right) dx$$

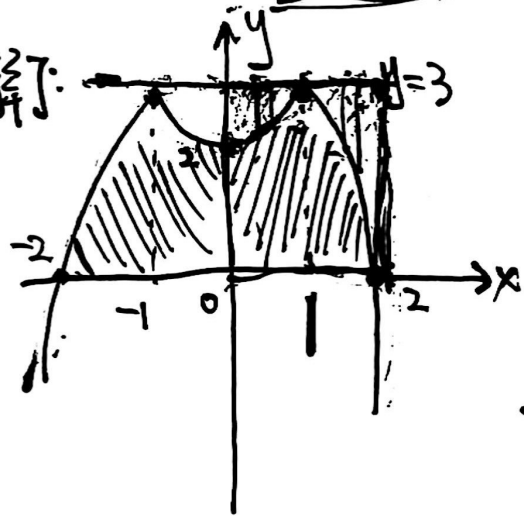
$$= 2\pi \cdot \int_0^1 (2-x) \cdot \sqrt{1-(x-1)^2} dx - 2\pi \cdot \int_0^1 (2-x) \cdot x dx$$

$$= \dots = \frac{\pi^2}{2} - \frac{2}{3}\pi \quad \checkmark$$

$$3 - [3 - |x^2 - 1|] = r = |x^2 - 1|$$

美题. 求曲线 $y = 3 - |x^2 - 1|$ 与 x 轴围成的封闭图形绕 $y=3$ 旋转所得的旋转体体积. (Ans: $\frac{448}{15}\pi$)

[解]:



$$V = 2V_{\text{壳}} = 2 \cdot (V_{\text{大}} - V_{\text{小}}) = 2 \cdot (V_{\text{大}} - V_1 - V_2)$$

其中 $V_{\text{大}}$ 是圆柱体体积. $V_{\text{大}} = \pi \cdot 3^2 \cdot 2 = 18\pi$

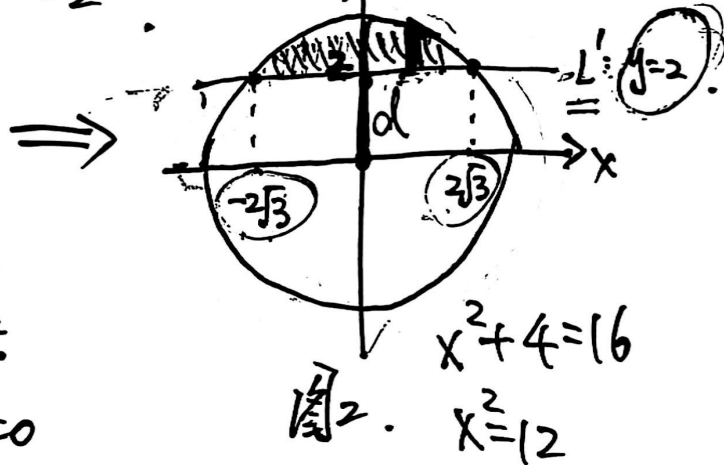
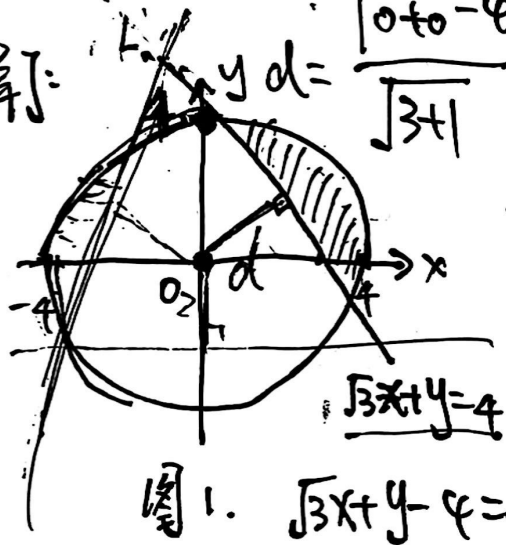
$$\begin{cases} V_1 = \int_0^1 \pi r^2 dx = \pi \int_0^1 (x^2 - 1)^2 dx \\ V_2 = \int_1^2 \pi r^2 dx = \pi \int_1^2 (x^2 - 1)^2 dx \end{cases} \Rightarrow V_1 + V_2 = \pi \int_0^2 (x^2 - 1)^2 dx = \dots$$

例9. 设圆周 $x^2+y^2=16$ 与直线 $L: \sqrt{3}x+y=4$ 围成的较小的那块弓形状的图形为 D .

求 D 绕直线 L 旋转一周生成的旋转体的表面积

→ 改为体积

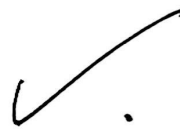
[解]: $d = \frac{|0+0-4|}{\sqrt{3+1}} = 2$



$$V = \pi \int_{-2\sqrt{3}}^{2\sqrt{3}} (16 - x^2 - 2)^2 dx$$

$$\stackrel{x=4\sin t}{=} 2\pi \int_0^{\frac{\pi}{3}} (4\cos t - 2)^2 4\cos t dt$$

$$= \dots = 12\sqrt{3}\pi \left(\frac{16}{3} \cdot \pi^2 \right)$$



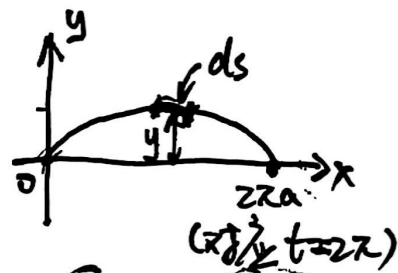
[注]: 若求表面积, 也可以用图2. 道理相同. (用弧长 ds 乘以走过的路, 就是扫出来的面积)

$$S = \int_{-2\sqrt{3}}^{2\sqrt{3}} 2\pi \cdot (\underbrace{\sqrt{16-x^2}-2}_{\text{走过的路}}) \cdot \underbrace{\sqrt{1+\frac{x^2}{16-x^2}}}_{ds} dx = \dots = 32 \cdot \left(\sqrt{3} - \frac{\pi}{3} \right) \cdot (\pi)$$

$$ds = \sqrt{1+(y')^2}, \quad y = \sqrt{16-x^2}$$

考法三. 平面曲线的弧长和旋转曲面的面积. (数一. 数二=).

例10. 摆线 $L: \begin{cases} x = a(t - \sin t) \\ y = a(1 - \cos t) \end{cases} \quad (0 \leq t \leq 2\pi, a > 0).$ 求:



(1) 这段弧的长度. (2) 这段弧绕 x 轴旋转一周所成的旋转曲面的面积.

(3) ----- y 轴 -----

$$ds = \sqrt{1 + y'^2} dx$$

$$\sqrt{1 \pm \cos t}$$

[解]: (1) $L = \int_L \overset{ds}{\sqrt{(dx)^2 + (dy)^2}} = \int_0^{2\pi} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt = \int_0^{2\pi} \sqrt{a^2(1 - \cos t)^2 + a^2 \sin^2 t} dt$

$$= a \int_0^{2\pi} \sqrt{2 - 2\cos t} dt = a \int_0^{2\pi} \sqrt{2 \cdot 2\sin^2 \frac{t}{2}} dt = 2a \int_0^{2\pi} \sin \frac{t}{2} dt = 8a.$$

(2) $S_x = \int_L (2\pi y) ds = \int_0^{2\pi} 2\pi \cdot a(1 - \cos t) \cdot \sqrt{a^2(1 - \cos t)^2 + a^2 \sin^2 t} dt = 8\pi a^2 \int_0^{2\pi} \sin^3 \frac{t}{2} dt = \frac{64}{3} \pi a^2$

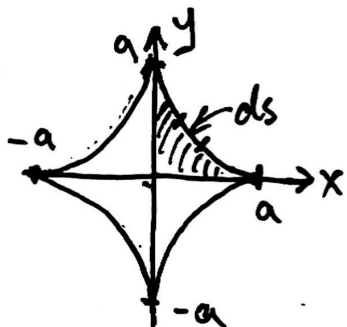
(3) $S_y = \int_L (2\pi x) ds = \int_0^{2\pi} 2\pi \cdot a(t - \sin t) \cdot \sqrt{a^2(1 - \cos t)^2 + a^2 \sin^2 t} dt = 4\pi a^2 \int_0^{2\pi} (t - \sin t) \cdot \sin \frac{t}{2} dt$

$$= \dots = 16\pi^2 a^2.$$

习题. 星形线 $L: \begin{cases} x = a \cos^3 t \\ y = a \sin^3 t \end{cases} (a > 0)$. 求:

(1) L 围成的面积. (2) L 的弧长. (3). L 绕着 y 轴旋转一周围成的旋转体体积和侧面积.

解:



$$\begin{aligned} (1). \text{面积 } A &= 4A_1 = 4 \cdot \int_0^a y dx = 4 \cdot \int_0^{\frac{\pi}{2}} a \sin^3 t \cdot a \cdot 3 \cos^2 t \cdot (-\sin t) dt \quad (\text{换元}) \\ &= 12a^2 \int_0^{\frac{\pi}{2}} \sin^4 t \cos^2 t dt = \frac{3}{8} \pi a^2. \\ &= 4 \cdot \int_0^{\frac{\pi}{2}} 3a \sin t \cos t dt = 6a^2. \end{aligned}$$

$$(2). \text{弧长 } L: L = 4L_1 = 4 \cdot \int_0^{\frac{\pi}{2}} \sqrt{[x'(t)]^2 + [y'(t)]^2} dt = 4 \cdot \int_0^{\frac{\pi}{2}} \sqrt{[a \cdot 3 \cos^2 t \cdot (-\sin t)]^2 + [a \cdot 3 \sin^2 t \cdot \cos t]^2} dt$$

$$\begin{aligned} (3). \underline{V_x} = 2V_{\text{右}} &= 2 \cdot \int_0^a \pi y^2 dx = 2 \cdot \int_0^{\frac{\pi}{2}} \pi (a \sin^3 t)^2 \cdot a \cdot 3 \cos^2 t \cdot (-\sin t) dt = 6\pi a^3 \int_0^{\frac{\pi}{2}} \sin^7 t \cos^2 t dt \\ &= \dots = \frac{32}{105} \pi a^3. \end{aligned}$$

$$\begin{aligned} \text{侧面积: } S &= 2S_{\text{右}} = 2 \cdot \int_0^{\frac{\pi}{2}} (2\pi y) \cdot \sqrt{[x'(t)]^2 + [y'(t)]^2} dt = 2 \cdot \int_0^{\frac{\pi}{2}} 2\pi \cdot a \sin^3 t \cdot 3a \sin t \cos t dt = 12\pi a^2 \int_0^{\frac{\pi}{2}} \sin^4 t \cos t dt \\ &= \frac{12}{5} \pi a^2. \end{aligned}$$